HOWARD WOON HAO ZHE

BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) UNIVERSITY OF MALAYA

Address: Kajang, Selangor | Phone: 016-237-3980 | Email: howardwoonhz06@gmail.com |
GitHub: HowardWoon | LinkedIn: Howard Woon Hao Zhe | Portfolio: https://howard-woon-hz-06-portfolio.vercel.app/

SUMMARY

Driven Software Engineering student (CGPA 4.00) specializing in full-stack development, AI/ML integration, and IoT solutions. Proven ability to build and deploy scalable applications under pressure, demonstrated through multiple award-winning hackathon projects across React, Node.js, and Python. Leveraging a background in corporate finance, brings rigorous analytical problem-solving and a data-driven approach to software architecture. Seeking a part time or IT related internship to accelerate academic development and contribute to engineering high-impact, production-ready systems that drive digital innovation and connectivity.

TECHNICAL SKILLS

Languages: Java, Python

Frameworks & Tools: React, Node.js, Firebase, Git/GitHub, RESTful APIs

AI / Machine Learning: Gemini API, Ollama (Local LLM Deployment), Generative AI

Core Concepts: Data Structures, Version Control, RESTful API Integration, Computer System and Architecture, Computer System and Organization, Machine Learning, Network Technology Foundation, Human Computer Interaction

CORE COMPETENCIES

Assistant Finance Executive — *Kraiburg TPE Technology (M) Sdn. Bhd*. **Jun 2025 - Sep 2025**

- Managed high-volume financial data pipelines, implementing strict verification protocols to ensure data integrity and transaction accuracy
- Applied analytical problem-solving to execute complex financial reconciliations, effectively "debugging" ledger discrepancies within enterprise reporting systems
- Processed vendor payments and employee expense claims in accordance with company compliance policy

Finance Intern — *Kraiburg TPE Technology (M) Sdn. Bhd*. **Mar 2024 - Jun 2024**

- Supported budget monitoring and expenditure tracking to improve financial planning accuracy
- Gained early exposure to enterprise data systems and structured reporting workflows

TECHNICAL PROJECTS

CATFISH.AI — Fraud Detection ML System **Machine Learning Project**

- Built a 6-model soft-voting ensemble (Decision Tree, MLP, SVM, Logistic Regression, KMeans, GMM) on a 50,000-row dataset
- Engineered a robust data pipeline using SMOTE-Tomek for class balancing and PCA for feature optimization; deployed the predictive model via a Flask REST API hosted on Vercel

Sensor X Sensei — Smart Lecture Hall Energy System **UM Technothon 2026 Top 15 Finalist**

- Developed an end-to-end IoT solution leveraging ESP32 microcontrollers and multi-sensor data fusion (NFC, PIR, TOF) to power a real-time occupancy dashboard
- Automated kWh/CO2 calculations using Malaysia's national grid emission factor

Slotify — Spring Boot Parking Management System **Data Structures Project**

- Engineered a Spring Boot backend implementing 7 hand-built data structures (doubly linked list, queue, stack, min-heap, weighted graph, HashMap, AVL tree) unified into a single end-to-end parking pipeline
- Implemented Dijkstra's shortest-path routing and a real-time interactive dashboard visualizing live heap re-heapification, AVL traversal, and O(1) hash table lookups

EDUCATION

Bachelor of Computer Science (Software Engineering) **Oct 2025 - Mar 2029**
University of Malaya | CGPA: 4.00

Physical Sciences Program **2024 – 2025**
Negeri Sembilan Matriculation College | CGPA: 4.00 | MUET BAND: 4.5

SPM (Sijil Pelajaran Malaysia) (Pure Science and Computer Science Program) **2019 – 2023**
SMJK Yu Hua Kajang | Result: 8A+, 1A, 2A

LEADERSHIP & ACTIVITIES

Finance Lead **Jun 2026 – Present**
Persatuan Komputer Universiti Malaya (PEKOM)

- Lead the financial architecture and resource management for Universiti Malaya's flagship technology community, driving the annual fiscal strategy to sustain student-led tech initiatives, hackathons, and professional development programs
- Architected and oversee the organization's financial portfolio, applying strict data verification protocols to maintain ledger accuracy and eliminate transaction discrepancies across all club operations
- Partner cross-functionally with the Sponsorship and PR departments to secure funding from corporate tech sponsors, using data-driven budget models to maximize student value and operational scale
- Spearheaded the transition from manual accounting to an automated digital claims pipeline, eliminating paperwork bottlenecks and scaling operational efficiency

Treasurer **Feb 2026 – Jun 2026**
MYTECH Career Fair 2026

- Managed a RM50,200 budget, secured 30 corporate sponsors and RM46,200 in revenue; enforced an 81.5% spending cap to deliver a RM9,272.90 surplus

AWARDS & CERTIFICATIONS

- **2nd Place** — AutoPilot Asia Hackathon 2026 (Sales Intelligence Track)
- **Top 5 Finalist & Consolation Prize —** DOSM Datathon 2026 (National Data Innovation Competition), Department of Statistics Malaysia
- **4th Place** — International Junior War Cry Competition 2022
- **Gold Medal —** National PPAL 4.0 Innovation Day
- **3rd Place —** MUBA Blockchain Hackathon 2026 (Payments and Stablecoins Track)
- **Champion —** Smart Water Utilities Track, Water Nexus Symposium 2026
- **Gold Medal —** Chemcreative-Innovation (Digital Learning)
- **Silver Medal** — Symposium for the Peer Assisted Learning Program, KPM Matriculation Division